

ReflCtrl: Controlling LLM Reflection via Representation Engineering

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Abstract

Large language models (LLMs) with Chain-of-Thought (CoT) reasoning have achieved strong performance across diverse tasks, including mathematics, coding, and general reasoning. A distinctive ability of these reasoning models is **self-reflection**: the ability to review and revise previous reasoning steps. While self-reflection enhances the reasoning performance, it also increases inference cost. In this work, we study self-reflection through the lens of **representation engineering**. We segment model’s reasoning into steps, identify the steps corresponding to reflection, and extract a reflection direction in the latent space that governs this behavior. Using this direction, we propose a stepwise steering method that can control reflection frequency. We call our framework ReflCtrl. Our experiments show that: (1) for many cases the reflections are redundant, especially in stronger models (in our experiment, we can save up to 33.6% of reasoning tokens while preserving the performance), and (2) model’s reflection behavior is highly correlated with internal uncertainty signal, implying self-reflection may be controlled by model’s uncertainty.

1 Introduction

Large language models (LLMs) have shown great success in many reasoning-related tasks, including math, coding, and general reasoning. A common technique for enhancing LLM reasoning is Chain-of-Thought (CoT) prompting [Wei et al., 2022], which asks the model to decompose the reasoning process into intermediate steps. Recently, a new class of models has been trained to develop **native reasoning ability**, such as OpenAI’s o1 [OpenAI, 2024] and DeepSeek-r1 [DeepSeekAI et al., 2025]. They can automatically generate reasoning steps before providing a response, even without being prompted to do so.

Notably, these reasoning models develop the ability to **self-reflect**, i.e. rethink their previous reasoning during training. This is described by DeepSeek-R1 [DeepSeekAI et al., 2025] as the “aha moment”. The self-reflection ability is a key difference between reasoning models and their non-reasoning counterparts, and is widely believed to contribute to improved reasoning ability. Additionally, it is also a costly component in inference: our empirical study finds self-reflection consumes 25-30% of total reasoning tokens.

Despite its potential importance, the underlying mechanism of self-reflection is not yet well understood. In this work, we take the first step to investigate this phenomenon through the lens of representation engineering [Zou et al., 2023], focusing on two central research questions:

RQ1: When does the model initiate reflection during its reasoning process?

RQ2: How does reflection influence the model’s reasoning performance?

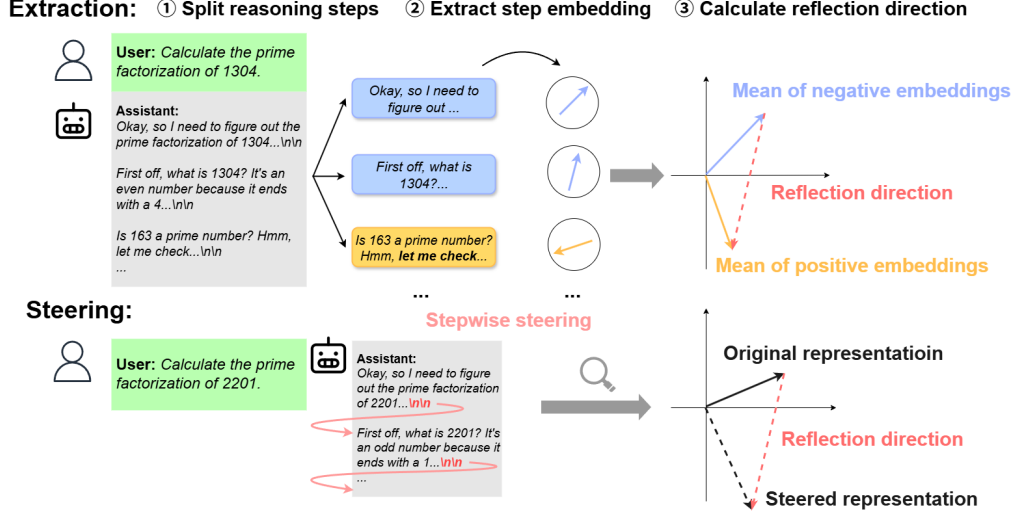


Figure 1: **Overview of the proposed ReflCtrl framework.** The model’s reasoning is first segmented into steps, then reflection-related steps are identified through keywords. Finally, a reflection direction is extracted by calculating the mean difference in the latent space. This direction can be used to steer the model’s self-reflection behavior via the proposed stepwise steering method, enabling control over reflection frequency and inference cost.

To answer these questions, we propose a novel method to identify the **reflection direction** in the model’s latent representation space. Our steering experiments demonstrate that this direction can effectively control the number of reflections during reasoning. Our empirical analysis further reveals that in many cases, the model’s reflections are redundant, offering an opportunity to reduce computational cost without sacrificing accuracy. Our contributions can be summarized as follows:

1. We identify a reflection direction in the model’s representation space that controls self-reflection, enabling us to steer model’s reflection behavior according to user’s intention.
2. We answer **RQ1** by connecting the model’s reflection direction to model’s internal uncertainty. In Sec. 3.5, we show that model’s activation along reflection direction can effectively predict answer’s correctness. This implies model’s reflection behavior may be controlled by internal uncertainty measurement.
3. Utilizing the reflection directions that we discovered, we steer model’s reflection to answer **RQ2**: Empirical results suggest in many cases model’s self-reflection could be redundant. Further, we design a novel stepwise steering method to address reflection redundancy. This new method reduces inference cost by up to 33.6%, while preserving reasoning performance.

2 Related works

Reasoning LLMs Motivated by the success of Chain-of-Thought reasoning, several models have been trained to enhance native reasoning capability by generating thinking steps. OpenAI’s o1 [OpenAI, 2024] leverages reinforcement learning to deliberate its thinking during inference. DeepSeekAI et al. [2025] introduces a more cost-efficient training method with the Grouped Relative Policy Optimization (GRPO) algorithm, as well as its distilled variants (DeepSeek-r1-distilled) that equip smaller models with thinking ability. QwQ-32b [Team, 2024] is a medium-sized reasoning model that achieves competitive performance with DeepSeek-r1 and o1. In this work, we focus on QwQ-32B [Team, 2024] and DeepSeek-r1-distill series, as they are open-sourced, allowing us to apply representation engineering techniques.

Representation engineering on LLMs While modern LLMs demonstrate remarkable capabilities, their internal mechanism is still not fully understood. Various efforts aim to make models more steerable [Srivastava et al., 2024, Sun et al., 2024, Kulkarni et al., 2025]: among these, representation

engineering [Zou et al., 2023, Bartoszcze et al., 2025] offers a principled way to analyze and guide model behavior by directly manipulating their internal representations. Zou et al. [2023] shows that representation engineering can be applied to multiple safety-related aspects by reading and editing model’s internal representation. With the rise of reasoning models, representation engineering methods specialized for these models have emerged: ThinkEdit [Sun et al., 2025b] identifies a set of neurons controlling “short-thinking” and mitigates it via weight editing. Wang et al. [2025b] identifies special experts that coordinate reasoning and improves models’ reasoning performance with a training-free method called RICE. Li et al. [2025] achieves machine unlearning via activation steering. Sun et al. [2025a] design LLMs to natively support interpretable concept steering. In contrast, our work focuses on the **reflection behavior** of reasoning LLMs, which is an interesting reasoning pattern introduced in reinforcement learning but not yet systematically investigated from a representation engineering perspective.

Self-reflection. DeepSeekAI et al. [2025] report that models learn to self-reflect autonomously, described as the “aha moment”. Yang et al. [2025] examine this phenomenon by comparing reasoning models with their non-reasoning counterparts in terms of linguistic patterns and description of uncertainty. Wang et al. [2025a] propose reducing excessive reflection by suppressing corresponding tokens to reduce models’ overthinking. In this work, we adopt a representation engineering perspective, revealing that models’ reflection is correlated with its internal uncertainty representation and can be directly controlled via our proposed method.

3 ReflCtrl: Probing and steering self-reflection

In this section, we investigate reflection behavior in reasoning models through the lens of representation engineering. We start by identifying reflection steps in the model’s reasoning, then extract a reflection direction in the latent space, and finally use this direction to steer the model’s behavior.

3.1 Background

Reasoning LLMs are built upon the Transformer decoder architecture [Vaswani et al., 2017], which stacks multiple identical layers. Each decoder layer l processes the hidden representation $z_l \in \mathbb{R}^d$. It consists of two major components: a self-attention block and a feed-forward MLP block. Formally, it can be written as:

$$\begin{aligned} \tilde{z}_l &= z_l + z_l^{\text{attn}}, \quad z_l^{\text{attn}} = \text{Attn}(\text{LN}(z_l)), \\ z_{l+1} &= \tilde{z}_l + z_l^{\text{mlp}}, \quad z_l^{\text{mlp}} = \text{MLP}(\text{LN}(z_l)). \end{aligned} \quad (1)$$

Here, $\text{LN}(\cdot)$ denotes layer normalization, $\text{Attn}(\cdot)$ is the self-attention block and $\text{MLP}(\cdot)$ is the feed-forward network. We denote $\tilde{z}_l$ as the intermediate state after the attention block.

3.2 Identify reflection behavior

Reasoning LLMs usually produce a long, multi-step thinking process. To facilitate our analysis of model’s reasoning, we first split the generated reasoning into thinking steps. We observe that such steps are naturally separated by the token sequence “\n\n” (an empty line) in most reasoning models, with each segment representing a coherent chunk of reasoning. Therefore, we treat each segment separated by “\n\n” as the smallest unit of analysis.

To identify reflection steps, we search for specific keywords within each step that mark the start of a new reflection, e.g., “Let me think”, “Wait”. While a reflection may span multiple steps, we identify it by detecting its initial step containing these keywords.

3.3 Extract reflection direction

With labeled reflection steps, we next compute the reflection direction in the latent space. For each step s at layer l , we extract all internal representations from the MLP and attention output of the first token, denoted as $z_l^{\{\text{attn}, \text{mlp}\}}(s)$. We use the first token because it captures the model’s internal state when reflection is initialized, allowing us to investigate the triggering mechanism of reflections. The reflection direction is then defined as the mean difference between reflection and non-reflection